Topology Final (2025 Fall, B 卷)

1. (12 points) Consider the mapping cone $C_f = S^1 \cup_f (CS^1)$, where $CS^1 = S^1 \times [0,1]/S^1 \times \{1\}$, $f : S^1 \times \{0\} \to S^1$ is given by $f(z,0) = z^2$. Compute $\pi_1(C_f)$, $H_0(C_f)$, $H_1(C_f)$ and $H_2(C_f)$.
2. (12 points) Let $S^1 = \{e^{i\theta}, 0 \le \theta < 2\pi\}$. Consider $X = S^1 \vee S^2$.
   (1) (4 points) Show that $\pi_1(X, x_0) \cong \mathbb{Z}$;
   (2) (8 points) Let $H = 2\mathbb{Z}$ be the subgroup of $\mathbb{Z}$, then there exists a covering space X_H and a covering map $p_H : X_H \to X$ corresponding to H. Describe X_H and p_H explicitly.
3. (12 points) Let $X = S^1 \times S^1$ be the torus and $A = S^1 \times \{x_0\} \subset X$ be a circle in X. Let $B = \{(y,z,0) \mid y^2 + z^2 = 1\} \subset S^2$ be a circle in S^2. Let Y be the identification space from $X \sqcup S^2$ by identifying the circle A with the circle B. Compute $H_n(Y)$ for $n \ge 0$.
4. (12 points) Let $X = S^1 \vee S^1 \vee S^3$, compute the Euler characteristic $\chi(X)$.
5. (14 points) Consider the following questions:
   (1) (7 points) Let $f : S^1 \to S^1$ with $\deg f = d$, show that f is homotopic to $\gamma : S^1 \to S^1$ defined by $\gamma(z) = z^d$;
   (2) (7 points) The map $g : S^1 \to S^1$ is defined by $g(e^{i\theta}) = e^{i(\theta + \sin\theta)}$, compute $\deg g$.
6. (12 points) Let $f : S^n \to \mathbb{R}^n$ be a continuous map satisfying $f(-x) = -f(x)$ for all $x \in S^n$. Show that there exists some $x_0 \in S^n$ such that $f(x_0) = 0$.
7. (14 points) Consider $f : \mathbb{R}\mathrm{P}^{2m+1} \to \mathbb{R}\mathrm{P}^{2m+1}$ be
$$f([x_0 : x_1 : \cdots : x_{2m+1}]) = [-x_0 : x_1 : \cdots : x_{2m+1}]$$
where $[x_0 : x_1, \cdots, x_{2m+1}]$ is the homogeneous coordinate of $\mathbb{R}\mathrm{P}^{2m+1}$ (recall: the homogeneous coordinate is $[x_0 : x_1 : \cdots : x_{2m+1}] = [\lambda x_0 : \lambda x_1 : \cdots : \lambda x_{2m+1}]$ for any $\lambda \neq 0$).
   (1) (12 points) Compute the trace of map $f_* : H_k(\mathbb{R}\mathrm{P}^{2m+1}; \mathbb{Q}) \to H_k(\mathbb{R}\mathrm{P}^{2m+1}; \mathbb{Q})$ for each k and then compute the Lefschetz number of f.
   (2) (2 points) Does f has any fixed point? Prove your answer.
8. (12 points) Let $f : S^n \to S^n$ be a map with degree 0.
   (1) (6 points) Show that there exist points $x, y \in S^n$ such that $f(x) = x$ and $f(y) = -y$;
   (2) (6 points) For $D^n \subset \mathbb{R}^n$ be the unit ball. If there exists a vector field v on D^n such that $v(x) \neq 0$ for all $x \in D^n$, show that there exists one point in ∂D^n such that $v(x)$ is radially outward and one point in ∂D^n such that $v(y)$ is radially inward.